❖ Comprehensions in Python

1. Comprehensions in Python

    - Comprehension is a readable way to create a new collection (List, Set, Dictionary) from an existing iterable using a single line of code.

    - Comprehension is a shortcut to create a new List, Set, or Dictionary using a single line instead of writing a loop.

2. Why Do We Use Comprehensions?
    - Without comprehension, we write

```
numbers = [1, 2, 3, 4, 5]

square = []

for i in numbers:
    square.append(i ** 2)

print(square)
```

- With List Comprehension:

```
numbers = [1, 2, 3, 4, 5]

square = [i ** 2 for i in numbers]

print(square)
```

- Less code
- More readable
- Faster

3. Types of Comprehensions
   1) List Comprehension
   2) Set Comprehension
   3) Dictionary Comprehension

4. Types of Operations with Comprehensions
   1) Transformation (Mapping)
   2) Filtering (Conditionals)
   3) Etc.

5. Syntax - [expression for item in iterable if condition]

   - **expression** → what to do with each item (transform, keep as is, etc.)
   - **iteration** → loop through each element in the iterable
   - **condition** → optional filter (decides whether to include the item)

❖ execution order :

1. **Iteration happens first** → Python takes one element from the iterable.
2. **Condition is checked** (if x % 2 == 0) → If True, continue; if False, skip this element.
3. **Expression is evaluated** (x**2) → The result is added to the new list.

1) List Comprehension
   - Use [ ]
   - Syntax :   [expression for variable in iterable Condition(optional)]

```
numbers = [1,2,3,4,5]

square = [i*i for i in numbers]

print(square)
```

```
numbers = [10,15,20,25,30]

even = [i for i in numbers if i % 2 == 0]

print(even)
```

2) Set Comprehension
   - Use {}
   - Syntax - {expression for variable in iterable   condition (opt)}

```
numbers = [1,2,2,3,4,4,5]

result = {i for i in numbers}

print(result)
```

3) Dictionary Comprehension
   - Use {key:value}
   - Syntax - {key:value for variable in iterable}

```
students = ["Ram","Shyam","Amit"]

length = {name:len(name) for name in students}

print(length)
```

```
square = {i:i*i for i in range(1,6)}

print(square)
```